

Robin Hur

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Education

University of Virginia, School of Engineering and Applied Science

Charlottesville, VA | 2024–2027

B.S in Mechanical Engineering, GPA: 3.72

- **Relevant coursework:** Statics, Strength of Materials, Thermodynamics, Fluid Mechanics, Dynamics, Material Science, Computational Methods, Mechatronics

Garnet Valley High School

Glen Mills, PA | 2020-2024

Class of 2024, GPA: 4.31/4.50

Skills

- **Design & Analysis:** SOLIDWORKS (CSWA), Autodesk Fusion, GD&T, Python, MATLAB, Bambu Studio, Microsoft Office, FEA, Fabrication & Rapid Prototyping, Technical Documentation
- **Languages:** English (native), Korean (fluent), French (intermediate)

Experience

Undergraduate Research Assistant, UVA Dept. of Material Science

Feb 2026 – Present

- Contributing to research on environmentally assisted cracking in aluminum alloys for aerospace applications, supporting work relevant to NASA and the F-35 program
- Developing a Python application integrating an Alicat MC 10SLPM D/5M mass flow controller and Monarch 6184-010-CAL THProbe to regulate nitrogen flow and temperature for environmentally assisted cracking experiments

Undergraduate Research Assistant, UVA Dept. of Chemical Engineering

Aug 2024 – Nov 2025

- Conducted all-atom molecular dynamics simulations using AMBER to study hydrogel formation at the atomic level
- Developed and managed 100+ simulation setup files with systematic organization workflows
- Automated Python analysis workflows to generate standardized CSV outputs with Git-based version control while generating and visualizing molecular trajectories in VMD to study structural evolution and interactions
- Produced publication-quality figures and supplementary data; contributed writing, data interpretation, and curated datasets to a manuscript under review

Research Outputs

- Publication: <https://doi.org/10.26434/chemrxiv-2026-8jxvq> (ChemRxiv preprint)

Jan 2026

Undergraduate Teacher Assistant, UVA Dept. of Electrical and Computer Engineering

Jan 2026 – Present

- Support students during and outside of the classroom by answering questions and reinforcing course material
- Assist with grading assignments, providing timely and constructive feedback
- Work closely with course instructors and Graduate head TAs to help prepare and manage course materials

Bridge Program Counselor, UVA School of Engineering and Applied Sciences

Jul – Aug 2025

- Mentored 38 first-year engineering students as part of a 6-member counselor team during a 3.5-week residential program, guiding academic, social, and personal development
- Served as a Resident Advisor, cultivating a supportive community, promoting student engagement
- Tutored pre-calculus to enhance foundational skills and prepare students for college

Projects

Mechatronics and Robotics Society, Mechanical Subteam

Aug 2025 – Present

- Designing and building a robot for the NASA Lunabotics competition
- Developed CAD models for a Lunabotics-style robot capable of manipulating lunar regolith
- Researched vibration frequency effects in granular material flow to optimize excavation subsystem performance

Potential Energy Propulsion Racer, Project Member

Nov 2025

- Collaborated on an engineering team to design and build a 3D-printed, potential-energy-powered racing vehicle
- Applied CAD modeling, mechanical transmission design, and prototyping to develop and optimize vehicle components
- Delivered final technical report and curated comprehensive inventory of CAD parts, assemblies, and design files